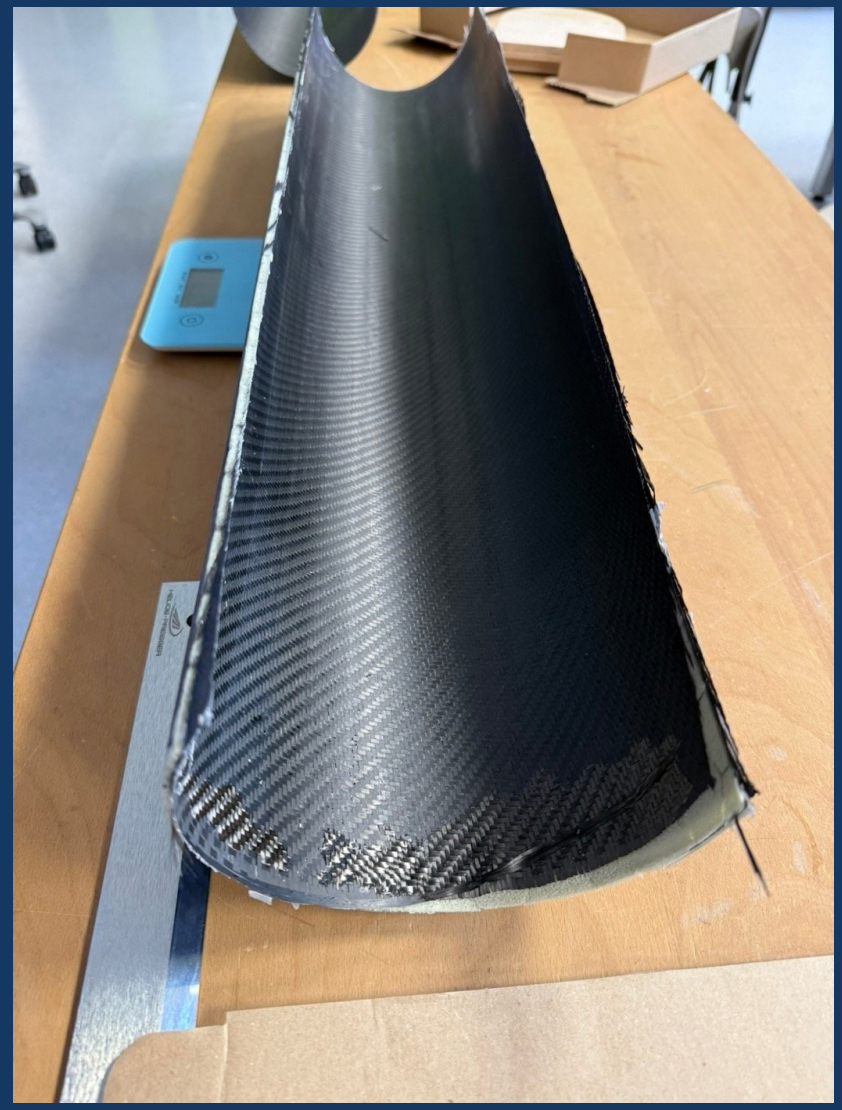
ERASMUS WORKSHOP · 18 NOVEMBER 2026

Building a carbon rotor blade

Sandwich construction with vacuum infusion

Windmobile · LTEtt



Five layers on a pipe

5 **Vacuum bag (tube film)** — width 86 cm
Everything slides inside; air pressure presses it together.

4 **Peel ply + flow mesh** — 87 × 32 cm
Spreads the resin. Peeled off after curing.

3 **Carbon** — 85 × 30 cm
Second structural layer.

2 **Core: 3DCore PET 100, 3 mm** — 85 × 30 cm
Foam with honeycomb pattern: stiff, light, lets the resin flow.

1 **Carbon** — 85 × 30 cm
Structural layer. Visible side on the outside.



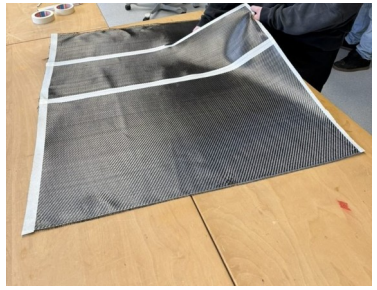
Peel ply (light) and flow mesh (yellow) from the roll



Second carbon layer goes onto the core

From dry fibres to blade

1 Cut



Tape the carbon, then cut carbon, core, peel ply + flow mesh

2 Lay up

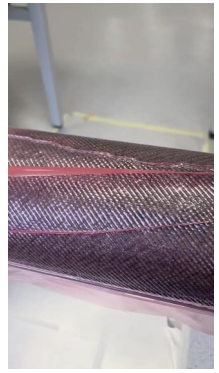


Carbon, core (spray adhesive), carbon, peel ply + flow mesh on the pipe

3 Bag + vacuum

Slide into the vacuum bag, attach and seal the hoses, pull the vacuum

4 Infuse



Mix 3 × (200 g resin + 60 g hardener), open the resin hose, watch the flow front

5 Cure

Infrared lamp speeds it up. Ready in the evening or the next day

Our two trial runs

TRIAL A

404 g

finished blade

Core: Lantor Soric SF

Resin + hardener: 400 g + 120 g

2 carbon layers



TRIAL B · workshop

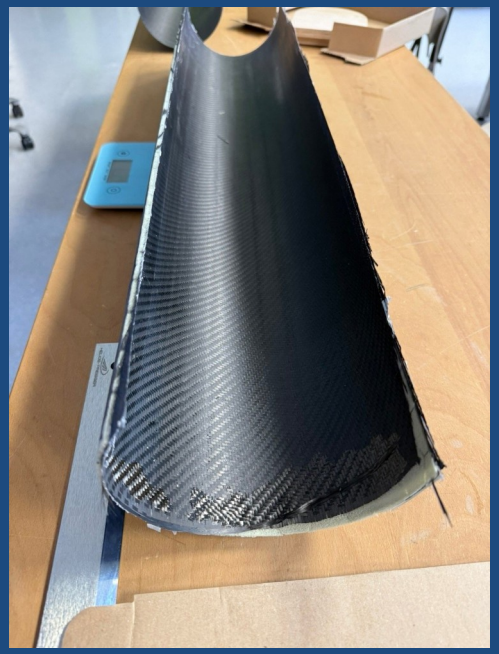
634 g

finished blade

Core: 3DCore PET 100, 3 mm

Resin + hardener: 600 g + 180 g

2 carbon layers



We build variant B: it is the only core material we have. Weigh your blade and compare.